Supplementary evidence and derivations for How Scoring Responses Change Attribution Evaluation in a Diffusion Policy

This supplement gives the proofs, experimental details and complete numerical tables supporting the main paper. Q scores negative squared action distance and N scores negative stabilized distance. The squared distance from the fixed reference action is r; replacing all features of the tested modality gives B, with other modalities fixed. Constructed examples establish mathematical possibilities, while the experimental tables describe their stated recorded populations. These two forms of evidence are kept separate.

Response identities and target-induced reversal

Normalized response difference

The identity proves how the normalized response changes while intervention predictions remain fixed. Exact self-reference means that the actual input reproduces the reference prediction, giving $r(x) = 0$. Assume $B > 0$. Let $\eta = \sqrt{\epsilon}$ be the stabilizer's square root, $b = \sqrt{B + \epsilon}$ the stabilized all-baseline distance and $z = \sqrt{r + \epsilon}$ the stabilized intermediate distance. Then $B = (b-\eta)(b+\eta)$ and

$$C_Q - C_N = \frac{b^2 - z^2}{B} - \frac{b - z}{b - \eta} = \frac{(z - \eta)(b - z)}{B}. \tag{S1}$$

For $\eta \le z \le b$, the product is nonnegative and maximized at $z = (\eta + b)/2$, giving

$$0 \le C_Q - C_N \le \frac{(b - \eta)^2}{4B} = \frac{b - \eta}{4(b + \eta)} \le \frac{1}{4}. \tag{S2}$$

For $z > b$, the difference is negative, so the $1/4$ upper bound holds for all $r \ge 0$. Nonnegative AUC weights sum to one, yielding $\Delta \le 1/4$ for every defined curve. On the nine-point schedule they are $(0.005, 0.025, 0.045, 0.075, 0.10, 0.15, 0.225, 0.25, 0.125)$. The endpoints contribute zero difference and have total weight 0.13, giving $\Delta \le 0.87/4 = 0.2175$. For positive ϵ and finite B, the fixed-B schedule bound $0.87(b - \eta)/[4(b + \eta)]$ is strictly smaller. Fully in-range curves additionally satisfy $\Delta \ge 0$, strictly if a point with positive AUC weight has $0 < r < B$. If $B = 0$, normalization is undefined. When the 1% and 5% language values coincide, dropping the 1% node changes their combined AUC weight by 0.02; the $1/4$ pointwise bound therefore bounds the paired change by $0.02/4 = 0.005$ for fully in-range calls when only that duplicate node

is removed. This also describes the realized-fraction change for the twenty-token schedule. For sixteen or twenty-five tokens, other realized fractions move as well, so 0.005 does not bound their total nominal-to-realized change.

Finite-stabilizer rank reversal

This example shows that changing only the attributed target can reverse the exact IG ranking. It is a constructed possibility, not an observed PickCube reversal. Let $g(x_1, x_2) = x_1^2 + \beta x_2$, with $\beta = 4/5$, $x' = (0, 0)$ and $x = (1, 1)$. Along the diagonal, the scalar action residual and its square are

$$\rho(\alpha) = (1 - \alpha)(1 + \alpha + \beta), \qquad r(\alpha, \alpha) = \rho(\alpha)^2.$$

Since $\nabla g = (2\alpha, \beta)$ there, differentiating the targets for $d_a = 1$ gives

$$\nabla Q = \rho(\alpha)(2\alpha, \beta), \qquad \nabla N = \frac{\rho(\alpha)}{\sqrt{\rho(\alpha)^2 + \epsilon}}(2\alpha, \beta). \tag{S3}$$

Polynomial integration yields

$$A^Q = \left(\frac{1}{2} + \frac{\beta}{3}, \frac{2\beta}{3} + \frac{\beta^2}{2}\right) = \left(\frac{23}{30}, \frac{64}{75}\right), \tag{S4}$$

so feature 2 leads by $13/150$. Norm attributions tend to $(1, \beta)$ as ϵ decreases to zero.

For the positive stabilizer, define $\psi(\rho) = 1 - \rho/\sqrt{\rho^2 + \epsilon} \geq 0$. Its derivative is $-\epsilon/(\rho^2 + \epsilon)^{3/2} < 0$, and $\rho(\alpha) \geq \beta(1 - \alpha)$. Therefore

$$\begin{aligned} \int_0^1 \psi(\rho(\alpha))\, d\alpha &\leq \int_0^1 \left[1 - \frac{\beta(1 - \alpha)}{\sqrt{\beta^2(1 - \alpha)^2 + \epsilon}}\right] d\alpha \\ &= 1 - \frac{\sqrt{\beta^2 + \epsilon} - \sqrt{\epsilon}}{\beta} \leq \frac{\sqrt{\epsilon}}{\beta}. \end{aligned} \tag{S5}$$

The identities $1 - A_1^N = \int_0^1 2\alpha\psi(\rho(\alpha))\, d\alpha$ and $\beta - A_2^N = \beta \int_0^1 \psi(\rho(\alpha))\, d\alpha$, together with $0 \leq 2\alpha \leq 2$ and $\epsilon = 10^{-12}$, give

$$0 \leq 1 - A_1^N \leq 2.5 \times 10^{-6}, \qquad 0 \leq \beta - A_2^N \leq 10^{-6}. \tag{S6}$$

Thus $A_1^N - A_2^N \geq 0.1999975$. Every coordinate is positive, so ordinary and absolute-coordinate rankings agree. Only the target changes. At the half-feature insertion point, feature 1 first gives normalized Q response about 0.80 versus 0.69 for feature 2 first, and N response about 0.56 versus 0.44. Deletion swaps these intermediate values. Hence feature 1 first scores better under both responses, despite exact Q-target IG ranking feature 2 first.

Finite-grid random-order calibration

This calculation shows why a random-order reference depends on the response, even when every input feature contributes equally to the action.

For $g(x) = \sum_{i=1}^{n} x_i$, an all-one input and zero baseline, every feature order gives residual distances k after k deletions and $n - k$ after k insertions. At $q = k/n$, normalized Q curves are $1 - q^2$ and $1 - (1 - q)^2$. Both have second derivative -2, so their trapezoidal areas underestimate their common integral $2/3$ by $1/(6n^2)$ on the full-prefix grid. Their identical curvature also gives identical quadrature errors on any common grid, including an asymmetric one. The nine-point schedule with $n = 100$ gives area 0.65976. Unstabilized N gives the lines $1 - q$ and q, both with trapezoidal area 0.5 on either schedule.

A common response change can reorder rankings

The response can also reverse which of two fixed feature orders has the better score. The following construction holds both orders and their intervention predictions fixed.

Take three features and insertion fractions $\{0, 1/3, 2/3, 1\}$, with $B = 1$. Let orders $(1, 2, 3)$ and $(3, 2, 1)$ give residual sequences $(1, 1/4, 1/4, 0)$ and $(1, 16/25, 0, 0)$, respectively. Their intermediate subsets are distinct, so these values can belong to one scalar policy with the same endpoints. Q insertion AUCs are $2/3$ and $31/50 = 0.620$; with $\epsilon = 0$, N AUCs are $1/2$ and $17/30 \approx 0.567$. Thus the better order reverses. The inequalities persist at $\epsilon = 10^{-12}$. This establishes possibility, not a measured comparison of the saved cohorts.

A smooth counterexample to integration-grid agreement

Unresolved coordinate variation

A smooth function can hide coordinate variation from every tested grid while its attribution sum remains exact. This example separates completeness and grid agreement from correct feature rankings.

For an integer $K > 0$, define

$$h(\alpha) = \frac{2}{5} \sin^2(\pi K \alpha), \tag{S7}$$

$$g(x_1, x_2) = \frac{2}{5} x_1 + \frac{3}{5} x_2 + (x_1 - x_2) h\left(\frac{x_1 + x_2}{2}\right), \tag{S8}$$

$$Q(x_1, x_2) = -\frac{1}{2}\big(g(x_1, x_2) - 1\big)^2. \tag{S9}$$

Take baseline $(0, 0)$, actual input $(1, 1)$, and fixed reference $g(1, 1) = 1$. The transverse term vanishes along (α, α), so $g(\alpha, \alpha) = \alpha$ and $Q(\alpha, \alpha) = -(1 - \alpha)^2/2$. Differentiating before restricting to that path gives

$$\nabla Q(\alpha, \alpha) = (1 - \alpha)\left(\frac{2}{5} + h(\alpha), \frac{3}{5} - h(\alpha)\right). \tag{S10}$$

Thus the oscillation transfers gradient between coordinates while leaving their sum unchanged.

Use the composite trapezoidal rule with m intervals. Whenever m divides K, every node $\alpha = j/m$ has $h(\alpha) = 0$. The sampled coordinate functions are then linear. Both the trapezoidal rule

and endpoint-inclusive average integrate a linear function exactly on any uniform grid, giving

$$A_m^{\text{trap}} = \left(\frac{1}{5}, \frac{3}{10}\right), \qquad \mathbf{1}^\mathsf{T} A_m^{\text{trap}} = \frac{1}{2} = Q(1,1) - Q(0,0). \tag{S11}$$

Here $\mathbf{1} = (1,1)^\mathsf{T}$ sums the coordinates. Every such grid has zero completeness residual and ranks feature 2 first, giving identical masks and perturbation curves under a deterministic ranking rule.

The exact coordinate integrals differ. Since $h(1-\alpha) = h(\alpha)$, substitution gives

$$\int_0^1 \alpha h(\alpha)\, d\alpha = \int_0^1 (1-\alpha) h(\alpha)\, d\alpha.$$

Adding these expressions and using $\int_0^1 h(\alpha)\, d\alpha = 1/5$ shows that either integral is $1/10$. Consequently,

$$A_{\text{exact}}^{Q} = \left(\frac{3}{10}, \frac{1}{5}\right). \tag{S12}$$

The exact order puts feature 1 first. Coarse errors $(-1/10, 1/10)$ cancel in completeness, but their L_1 norm, $1/5$, is 40% of the exact attribution's norm.

On the $2K$ grid, h alternates between zero and $2/5$, with trapezoidal integral $1/5$. Reflection symmetry gives area $1/10$ for $(1-\alpha)h$, recovering Equation S12. This recovery is construction-specific. For finitely many uniform budgets, a common multiple K hides every oscillation. Agreement cannot certify coordinate integrals without a regularity bound on unresolved variation.

The endpoint-inclusive average has the same coarse failure. When m divides K, the average of the sampled linear functions also yields Equation S11. On the $2K$ grid, only odd nodes contribute to $(1-\alpha)h$. Their sum is $\frac{2}{5}\sum_{\ell=0}^{K-1}(1-(2\ell+1)/(2K)) = K/5$. Dividing by the $2K+1$ nodes gives

$$A_{2K}^{\text{avg}} = \left(\frac{1}{5} + \frac{K}{5(2K+1)}, \frac{3}{10} - \frac{K}{5(2K+1)}\right). \tag{S13}$$

The first coordinate exceeds the second for $K > 1/2$, including every allowed integer. Coordinates remain inexact; exact $2K$ recovery belongs to the trapezoidal rule.

Exact IG and perturbation quality

Recovering the exact integral need not optimize a chosen insertion or deletion score. The same construction makes that distinction explicit.

Set $K = 128$, so $h(1/2) = 0$ and the whole-feature interventions give $g(1,0) = 2/5$ and $g(0,1) = 3/5$. Under the same normalized Q response and fractions $\{0, 1/2, 1\}$, feature-2-first insertion values are $(0, 0.84, 1)$ and deletion values are $(1, 0.64, 0)$. Their trapezoidal AUCs are 0.67 and 0.57. The exact-IG feature-1-first order instead gives insertion AUC 0.57 and deletion AUC 0.67. The inaccurate order therefore scores better by both conventional criteria. The transverse term alters coordinate gradients along the diagonal without altering these finite off-path intervention values. Exact path integration and perturbation-score optimization are distinct questions.

CPU fp32/fp64 checks at $K = 128$ verify trapezoidal coordinates, completeness, repeatability and responses: budgets 32, 64 and 128 alias; 256 recovers the analytic order to numerical tolerance. The endpoint-inclusive result is analytic only; these checks use no RDT data.

Endpoint transitions and tail bounds

The stabilized norm target has a narrow endpoint transition that explains the tested integration behavior. In the linear action example, write the remaining action distance as $t = 1 - \alpha$. The gradient magnitude is $t/\sqrt{t^2 + \epsilon}$, and the exact integral is $\sqrt{1 + \epsilon} - \sqrt{\epsilon}$. With $\epsilon = 10^{-12}$, grids of 64 through 512 intervals do not resolve the transition near $t = 0$. Assigning weight to its zero endpoint then produces approximately inverse-budget error. This describes the tested regime, not the asymptotic order for fixed positive epsilon. A sufficiently fine grid resolves the smooth transition and restores second-order trapezoidal convergence. Midpoint and Gauss-Legendre rules avoid endpoint nodes but still require accuracy checks for the particular integrand.

Some tail thresholds follow from the response identity and cannot be treated as independent observations. Put $R = \max r/B$ over interior nodes and $s = \sqrt{R}$. Assume $R > 1$, as required for a negative Δ. For an overshooting point, stabilization reduces the magnitude of the negative difference relative to its zero-stabilizer limit; the minimum is bounded below by $s(1 - s)$. Since the saved schedule's interior weights sum to 0.87, $\Delta \geq 0.87 s(1 - s)$. Consequently, for $a > 0$, $\Delta < -a$ requires

$$R > \left[\frac{1 + \sqrt{1 + 4a/0.87}}{2}\right]^2 .$$

For $a = 1$ and $a = 100$ the required ratios are approximately 2.83 and 126.2. The empirical findings are the actual residual ratios, small baseline distances and concentration in particular calls, not the fact that these necessary thresholds hold. The sixteen correlation groups comprise eight saved pretrained PickCube cases, two pooled task-trained vision cases and six seed-specific task-trained language cases. Their negative rank correlations mean that larger residual ratios accompany more negative Δ.

Experimental details and complete results

The following tables retain each population separately. The two saved pretrained PickCube runs use different ranking targets and language baselines; shared simulator seeds do not make them matched experiments. The saved task-trained PickCube primary sample pools seeds 42, 142 and 242, with 20 episodes each, respectively 320, 480 and 399 calls, and 9, 1 and 5 successes. The separate 334-call rerun is a sensitivity sample and is excluded from the 1,199-call primary total. Each saved additional task pools two eight-episode runs at 128 intervals. The 800 language curves used to check distinct StackCube 1%/5% responses comprise two 200-call runs and both intervention directions.

The offline diagnostic collection contains twelve completed contexts per input type and target from six episodes. The preserved bank used in the matched numerical campaign contains the same 21 available identifiers from that collection's 24 planned identifiers. It has four pretrained contexts each for PickCube, StackCube, PegInsertionSide and PickSingleYCB, two task-trained PickCube contexts and three task-trained StackCube contexts, from twelve source episodes overall. Only twelve pretrained contexts had complete offline diagnostic results at the collection's sealed cutoff. These counts describe different subsets and must not be combined as independent episodes. Small complete-case strata have at most two episodes. Nominal 95% percentile intervals contain the

known value in only about 9%–50% of the examined small-sample simulations; those strata support no population-level conclusion.

Saved analyses use the nominal percentage axis and bootstrap seed zero. Matched analyses use the actual fractions selected by stored masks and seed 930011, with nominal-axis sensitivity results retained. The endpoint grids share 513 gradient samples, allowing nested trapezoidal and equal-weight estimates through 512 intervals and midpoint estimates through 256. Original-estimator sums can depend on GPU summation order; the separate 32-bit-sum alternative remains recorded. Repeated saved IG sums can differ despite identical target changes, which is another reason endpoint agreement does not authenticate the missing original computation. Unstabilized RMS action distance normalizes to the same curve as unstabilized N and supplies no third independent normalized metric.

Saved response summaries

Table S1 retains the eight saved cases, including tails and uncertainty. Its intervals condition on the recorded episodes; repeated calls and shared extreme episodes do not constitute independent replications.

Table S1 | All eight saved response-change cases. Each row contains 750 calls in 30 episodes. Q/N identifies the ranking target. Brackets give descriptive 95% whole-episode bootstrap intervals, conditional on recorded episodes, not variation across training runs, checkpoints or tasks. Medians weight calls equally and means weight episodes equally. Positive means $\Delta > 0$; overshoot means $r > B$ at an interior sample. Both columns are percentages. These unmatched runs do not compare ranking quality. Three negative mean intervals include zero.

Target, input, direction	Median [interval]	Positive overshoot (%)	/ Mean [interval]	Minimum
Q vision insertion	0.1042 [0.0961, 0.112]	88.1 / 41.6	−17.1 [−33.8, −3.97]	−4,439
Q vision deletion	0.1588 [0.148, 0.169]	92.5 / 16.9	0.123 [0.106, 0.138]	−1.90
Q language insertion	0.1169 [0.0667, 0.137]	75.6 / 74.0	−0.182 [−0.405, −0.0277]	−71.8
Q language deletion	0.1189 [0.0912, 0.138]	88.5 / 37.5	−0.00947 [−0.102, 0.0631]	−23.8
N vision insertion	0.1041 [0.0947, 0.113]	90.0 / 38.9	−18.6 [−37.9, −3.17]	−5,243
N vision deletion	0.1590 [0.146, 0.169]	94.3 / 16.0	0.0184 [−0.204, 0.138]	−80.5
N language insertion	0.1279 [0.111, 0.145]	84.1 / 65.5	−0.0329 [−0.0963, 0.0266]	−13.3
N language deletion	0.1205 [0.0991, 0.131]	83.2 / 41.6	−1.37 [−4.21, 0.0780]	−1,055

Component identities and recovery status

Table S2 distinguishes information written at run time, recovered from dated records, inferred indirectly and unknown. A matching downloaded file history does not mean its hash was recorded during a saved run. Full hashes and machine captures remain in the research archive.

Table S2 | Component identity status for the main saved and new groups. Recorded means captured during that experiment; recovered means reconstructed from dated evidence; inferred means supported but not established; unknown means unavailable. Recovery does not imply a runtime hash. Full values and all 12 population inventories accompany the public source projection. Saved checkpoint and encoder files can be recovered, while most runtime versions remain unknown. New runtime entries include incomplete host captures and do not establish every host individually.

Component	Saved Q	Saved N	New 170M PickCube	New 1B PickCube	New 170M StackCube
model repository	recovered	recovered	recorded	recorded	recorded
model revision	recovered	recovered	recorded	recorded	recorded
model weight file and sha256	recovered	recovered	recorded	recorded	recorded
rdt upstream code commit	recovered	recovered	recorded	recorded	recorded
model config	recovered	recovered	recorded	recorded	recorded
vision encoder	recovered	recovered	recorded	recorded	recorded
vision preprocessor	recovered	recovered	inferred	inferred	inferred
language encoder or embedding source	inferred	recovered	recorded	recorded	recorded
language baseline	inferred	recovered	recorded	recorded	recorded
language positions	recovered	recovered	recorded	recorded	recorded
adaptor identity	recovered	recovered	recorded	recorded	recorded
simulator package versions	unknown	unknown	inferred	inferred	inferred
task	recorded	recorded	recorded	recorded	recorded
control mode	recovered	recovered	recorded	recorded	recorded
action subsampling	recovered	recovered	recorded	recorded	recorded
episode horizon	recorded	recorded	recorded	recorded	recorded
environment reset seeds	recorded	recorded	recorded	recorded	recorded
camera observation pipeline and image size	recovered	recovered	recorded	recorded	recorded
proprioception preprocessing	recovered	recovered	recorded	recorded	recorded
diffusion sampler	recovered	recovered	recorded	recorded	recorded
solver steps	recorded	recovered	recorded	recorded	recorded
noise policy	recovered	recovered	recorded	recorded	recorded

Component	Saved Q	Saved N	New 170M PickCube	New 1B PickCube	New 170M StackCube
precision dtype	recovered	recovered	recorded	recorded	recorded
runtime determinism settings	recovered	recovered	recorded	recorded	recorded
ig estimator	recovered	recovered	recorded	recorded	recorded
ig budget	inferred	recovered	recorded	recorded	recorded
ig target	recorded	recorded	recorded	recorded	recorded
nonfinite gradient handling	recovered	recovered	recorded	recorded	recorded
stored scoring response	recorded	recorded	recorded	recorded	recorded
intervention schedule and count rule	recovered	recovered	recorded	recorded	recorded
sw python	inferred	unknown	inferred	inferred	inferred
sw torch	inferred	recovered	inferred	inferred	inferred
sw numpy	unknown	unknown	inferred	inferred	inferred
sw diffusers	unknown	unknown	inferred	inferred	inferred
sw transformers	unknown	unknown	inferred	inferred	inferred
sw mani skill	unknown	unknown	inferred	inferred	inferred
sw sapien	unknown	unknown	inferred	inferred	inferred
hardware	inferred	recovered	recorded	recorded	recorded
project code version	inferred	inferred	recorded	recorded	recorded
preserved inputs	recorded	recorded	recorded	recorded	recorded

Identity status, continued: additional saved and preserved populations. The source projection retains full values and evidence status for every field.

Component	Saved 1B, seed 42	Saved 1B, seeds 142/242	Saved 1B re-run	Additional tasks, seed 142
model repository	recovered	inferred	inferred	recovered
model revision	recovered	inferred	inferred	recovered
model weight file and sha256	recovered	inferred	inferred	recovered
rdt upstream code commit	recovered	recovered	recovered	recovered
model config	recovered	recovered	recovered	recovered
vision encoder	recovered	recovered	recovered	recovered
vision preprocessor	recovered	recovered	recovered	recovered
language encoder or embedding source	recovered	recovered	recovered	recovered
language baseline	recovered	recovered	recovered	recovered

Component	Saved 1B, seed 42	Saved 1B, seeds 142/242	Saved 1B re-run	Additional tasks, seed 142
language positions	recovered	recovered	recovered	recovered
adaptor identity	recovered	inferred	inferred	recovered
simulator package versions	unknown	unknown	unknown	unknown
task	recorded	recorded	recorded	recorded
control mode	recovered	recovered	recovered	recovered
action subsampling	recovered	recovered	recovered	recovered
episode horizon	recorded	recorded	recorded	recorded
environment reset seeds	recorded	recorded	recorded	recorded
camera observation pipeline and image size	recovered	recovered	recovered	recovered
proprioception preprocessing	recovered	recovered	recovered	recovered
diffusion sampler	recovered	recovered	recovered	recovered
solver steps	recovered	recorded	recorded	recovered
noise policy	recovered	recovered	recovered	recovered
precision dtype	recovered	recovered	recovered	recovered
runtime determinism settings	recovered	recovered	recovered	recovered
ig estimator	recovered	recovered	recovered	recovered
ig budget	recovered	recovered	inferred	inferred
ig target	recovered	recovered	inferred	recovered
nonfinite gradient handling	recovered	recovered	recovered	recovered
stored scoring response	recorded	recorded	recorded	recorded
intervention schedule and count rule	recovered	recovered	recovered	recovered
sw python	unknown	inferred	unknown	unknown
sw torch	recovered	recovered	recovered	recovered
sw numpy	unknown	unknown	unknown	unknown
sw diffusers	unknown	unknown	recovered	unknown
sw transformers	unknown	unknown	recovered	unknown
sw mani skill	unknown	unknown	unknown	unknown
sw sapien	unknown	unknown	unknown	unknown
hardware	recovered	recovered	recovered	recovered
project code version	inferred	inferred	inferred	inferred
preserved inputs	recorded	recorded	recorded	recorded

Identity status, continued: additional saved and preserved populations. The source projection retains full values and evidence status for every field.

Component	Additional tasks, seed 42	Four-task collection	Preserved-context checks
model repository	recovered	recorded	recorded
model revision	recovered	recorded	recorded
model weight file and sha256	recovered	recorded	recorded
rdt upstream code commit	recovered	recorded	recorded
model config	recovered	recorded	recorded
vision encoder	recovered	recorded	recorded
vision preprocessor	recovered	recovered	recorded
language encoder or embedding source	recovered	recorded	recorded
language baseline	recovered	recorded	recorded
language positions	recovered	recorded	recorded
adaptor identity	recovered	recorded	recorded
simulator package versions	unknown	recorded	recorded
task	recorded	recorded	recorded
control mode	recovered	recorded	recorded
action subsampling	recovered	recorded	recorded
episode horizon	recorded	recorded	recorded
environment reset seeds	recorded	recorded	recorded
camera observation pipeline and image size	recovered	recorded	recorded
proprioception preprocessing	recovered	recorded	recorded
diffusion sampler	recovered	recorded	recorded
solver steps	recorded	recorded	recorded
noise policy	recovered	recorded	recorded
precision dtype	recovered	recorded	recorded
runtime determinism settings	recovered	recorded	recorded
ig estimator	recovered	recorded	recorded
ig budget	recorded	recorded	recorded
ig target	recovered	recorded	recorded
nonfinite gradient handling	recovered	recorded	recorded
stored scoring response	recorded	recorded	recorded
intervention schedule and count rule	recovered	recorded	recorded
sw python	unknown	recovered	recorded
sw torch	recovered	recorded	recorded

Component	Additional tasks, seed 42	Four-task collection	Preserved-context checks
sw numpy	unknown	recorded	recorded
sw diffusers	unknown	recorded	recorded
sw transformers	unknown	recorded	recorded
sw mani skill	unknown	recorded	recorded
sw sapien	unknown	recorded	recorded
hardware	recovered	recovered	recorded
project code version	inferred	recorded	recorded
preserved inputs	recorded	recorded	recorded

Values for the new experiment. Revisions and digests are 12-character prefixes. The saved 170M checkpoint and SigLIP revision were recovered as the same files, without runtime hashes. Complete per-population values remain in the source projection.

Component	Value	Status
RDT-170M checkpoint	rdt-170m; revision 8aa386cac3bb; SHA-256 2e87fccee029	recorded
RDT-1B checkpoint	maniskill-model/rdt; revision 9622afab2b7c; SHA-256 3fa89cbf7bbb	recorded
Vision encoder	SigLIP so400m-patch14-384; revision 9fdffc58afc9	recorded
Language encoder	T5-v1.1-XXL; revision 3db67ab1af98; preserved bf16 task embeddings	recorded
Code	RDT upstream cd79363a1387; study source d10de3d61747	recorded
Sampler	DPMSolverMultistepScheduler; 5 steps; explicit-noise adapter v1	recorded
Runtime	Python 3.12.3; torch 2.8.0+cu128; diffusers 0.27.2; transformers 4.41.0	inferred
Simulator	ManiSkill 3.0.1; SAPIEN 3.0.3; gymnasium 1.3.0	inferred
Control	pd_joint_pos; 20 Hz control / 100 Hz simulation; 400-step horizon	recorded
Determinism	Deterministic algorithms; no TF32; cuDNN deterministic; benchmark off	recorded
Image preprocessing	SigLIP: 384 × 384; resample 3; scale 1/255; mean/std 0.5	inferred
Adaptors	Checkpoint-contained: image/language mlp2x_gelu; state mlp3x_gelu; hashes archived	recorded

All matched method comparisons

Table S3 includes the twelve image-patch cells and eight pretrained-PickCube language/state cells. Best-versus-second comparisons are selected on the same data, adjusted within each four-cell family and interpreted as exploratory. Individual method intervals do not test a paired difference between two methods. Its continuation reports twelve saved image-patch deletion replication contrasts for StackCube and task-trained PickCube, with descriptive percentile and bias-corrected and accelerated (BCa) 95% intervals.

Table S3 | All 20 new-experiment method comparisons. The winner maximizes the raw episode-mean advantage. The gap is winner minus runner-up, with the registered interval adjusted across four response/direction cells within each group and input type. These selected contrasts do not correct for choosing the winner on the same data, nor across all groups. Three intervals exclude zero: one vision cell and two language cells.

Group and input	Scoring / direction	Raw winner	Mean advantage	Gap [adjusted interval]	Excludes zero
PickCube 1B vision	Q deletion	Input difference	−0.572	2.53 [−1.02, 12.2]	No
PickCube 1B vision	Q insertion	IG-Q	5.17	0.000714 [−0.000990, 0.00401]	No
PickCube 1B vision	N deletion	Path-gradient N	0.355	0.00840 [−0.0667, 0.0461]	No
PickCube 1B vision	N insertion	IG-Q	0.503	0.000276 [−0.000717, 0.00183]	No
PickCube 170M language	Q deletion	Path-gradient Q	−0.255	0.0238 [−0.0383, 0.102]	No
PickCube 170M language	Q insertion	Path-gradient N	−0.0531	0.0959 [0.0181, 0.215]	Yes
PickCube 170M language	N deletion	IG-Q	0.0205	0.000507 [−0.00132, 0.00258]	No
PickCube 170M language	N insertion	Path-gradient N	0.0603	0.0405 [0.0158, 0.0709]	Yes
PickCube 170M state	Q deletion	Path-gradient Q	0.278	0.00124 [−0.0704, 0.102]	No
PickCube 170M state	Q insertion	IG-N	0.306	0.00711 [−0.0766, 0.106]	No
PickCube 170M state	N deletion	Input difference	0.256	0.00479 [−0.00612, 0.0195]	No
PickCube 170M state	N insertion	Input difference	0.303	0.0117 [−0.0146, 0.0380]	No

Group and input	Scoring / direction	Raw winner	Mean advantage	Gap [adjusted interval]	Excludes zero
PickCube 170M vision	Q deletion	Path-gradient N	3.87	0.0998 [0.0181, 0.315]	Yes
PickCube 170M vision	Q insertion	Path-gradient N	−11.3	8.51 [−0.109, 23.8]	No
PickCube 170M vision	N deletion	Equal-weight IG-Q	0.476	0.0101 [−0.0149, 0.0606]	No
PickCube 170M vision	N insertion	Path-gradient N	−0.00388	0.0601 [−0.00149, 0.198]	No
StackCube 170M vision	Q deletion	Path-gradient N	0.559	0.00352 [−0.0159, 0.0285]	No
StackCube 170M vision	Q insertion	Path-gradient N	0.0872	0.00480 [−0.00250, 0.0174]	No
StackCube 170M vision	N deletion	Equal-weight IG-N	0.510	0.000862 [−0.000444, 0.00286]	No
StackCube 170M vision	N insertion	Path-gradient N	0.128	0.00297 [−0.00157, 0.00739]	No

Table S3 continued: saved replication image-patch deletion contrasts. IG-Q and IG-N use Q and N attribution targets; the response follows the slash. Advantage is random-order AUC minus method AUC. Each estimate subtracts the control advantage from the first method advantage; positive values favor the first method, negative values the control. These saved percentile and bias-corrected and accelerated (BCa) 95% intervals are descriptive, without the Bonferroni adjustment used for the primary family or a simultaneous guarantee. For task-trained PickCube, IG-N versus random under N excludes zero with the percentile interval but includes it with BCa. No episodes were excluded.

Group (episodes)	Contrast / response	Mean	Percentile 95% interval	BCa 95% interval
StackCube 170M (30)	IG-Q − random / Q	0.514	[0.439, 0.593]	[0.444, 0.600]
StackCube 170M (30)	IG-Q − random / N	0.483	[0.432, 0.537]	[0.436, 0.542]
StackCube 170M (30)	IG-N − random / Q	0.553	[0.470, 0.640]	[0.474, 0.647]
StackCube 170M (30)	IG-N − random / N	0.509	[0.451, 0.569]	[0.455, 0.573]

Group (episodes)	Contrast / response	Mean	Percentile 95% interval	BCa 95% interval
StackCube 170M (30)	IG-Q − IG-N / Q	−0.0397	[−0.0684, −0.0179]	[−0.0788, −0.0218]
StackCube 170M (30)	IG-Q − IG-N / N	−0.0254	[−0.0444, −0.0114]	[−0.0532, −0.0140]
PickCube 1B (40)	IG-Q − random / Q	−3.10	[−8.60, 0.249]	[−13.2, −0.241]
PickCube 1B (40)	IG-Q − random / N	0.347	[0.183, 0.459]	[0.0960, 0.439]
PickCube 1B (40)	IG-N − random / Q	−3.80	[−10.8, 0.294]	[−17.1, −0.232]
PickCube 1B (40)	IG-N − random / N	0.346	[0.130, 0.484]	[−0.0260, 0.460]
PickCube 1B (40)	IG-Q − IG-N / Q	0.698	[−0.0576, 2.19]	[−0.0492, 3.68]
PickCube 1B (40)	IG-Q − IG-N / N	0.000437	[−0.0321, 0.0546]	[−0.0275, 0.104]

Calibration and fixed decision rules

Table S4 reports scaled preferences, fixed-order controls and illustrative fixed-bar decisions. Subtracting the same random reference shifts every method equally within one response. Per-context scaling changes weights before episode averaging and is a different operation. Neither 0.40 nor 0.50 is a validated pass/fail criterion.

Table S4 | Scaled choices and fixed-order comparisons. Scaling is per context before episode averaging. Fixed orders use no learned attribution. The gap compares the best raw attribution method with the highest-mean fixed order; negative values favor that fixed order. Brackets retain the registered adjusted intervals. Mean comparisons can be tail-sensitive and do not establish a generally superior explanation.

Group and input	Scoring / direction	Scaled winner	Best fixed order	Gap [adjusted interval]
PickCube 1B vision	Q deletion	Path-gradient N	Raster	−1.89 [−9.45, 1.80]
PickCube 1B vision	Q insertion	Equal-weight IG-Q	Center-out	0.804 [−1.07, 3.76]
PickCube 1B vision	N deletion	Path-gradient N	Center-out	0.0440 [−0.274, 0.232]
PickCube 1B vision	N insertion	Equal-weight IG-N	Center-out	0.298 [0.217, 0.439]
PickCube 170M language	Q deletion	IG-Q	Reverse tokens	−1.30 [−3.29, −0.327]
PickCube 170M language	Q insertion	Path-gradient N	Reverse tokens	−0.372 [−1.53, 0.122]
PickCube 170M language	N deletion	IG-Q	Reverse tokens	−0.287 [−0.501, −0.132]
PickCube 170M language	N insertion	Path-gradient N	Reverse tokens	0.0263 [−0.109, 0.118]
PickCube 170M state	Q deletion	Input difference	Reverse joints	0.232 [0.155, 0.332]
PickCube 170M state	Q insertion	Input difference	Reverse joints	0.327 [0.198, 0.507]
PickCube 170M state	N deletion	Input difference	Reverse joints	0.192 [0.164, 0.221]
PickCube 170M state	N insertion	Input difference	Reverse joints	0.223 [0.181, 0.272]
PickCube 170M vision	Q deletion	IG-N	Raster	6.37 [−1.13, 26.2]
PickCube 170M vision	Q insertion	Path-gradient N	Raster	−18.3 [−60.9, 0.424]

Group and input	Scoring / direction	Scaled winner	Best fixed order	Gap [adjusted interval]
PickCube 170M vision	N deletion	IG-N	Reverse raster	0.257 [0.0924, 0.656]
PickCube 170M vision	N insertion	Path-gradient N	Reverse raster	−0.0393 [−0.561, 0.231]
StackCube 170M vision	Q deletion	Path-gradient N	Reverse raster	0.103 [0.0452, 0.150]
StackCube 170M vision	Q insertion	Path-gradient N	Reverse raster	0.0876 [0.0451, 0.130]
StackCube 170M vision	N deletion	Equal-weight IG-N	Reverse raster	0.124 [0.0910, 0.152]
StackCube 170M vision	N insertion	Path-gradient N	Reverse raster	0.112 [0.0884, 0.138]

Illustrative fixed-bar choices for vision deletion. Each cell lists all methods tied for the largest fraction of calls with AUC at or below the bar, and that fraction. These arbitrary bars are not validated decision rules. This criterion differs from the raw mean choice in four of six cells for either bar.

Group	Scoring	Bar 0.40	Bar 0.50
PickCube 170M	Q	IG-N (75.8%)	Equal-weight IG-N; IG-N (89.2%)
PickCube 170M	N	Equal-weight IG-N; IG-N; Equal-weight IG-Q; IG-Q (93.3%)	Equal-weight IG-N; IG-N; Input difference (98.3%)
PickCube 1B	Q	Path-gradient N (92.5%)	Equal-weight IG-N; IG-N; Path-gradient N (98.8%)
PickCube 1B	N	Equal-weight IG-N; IG-N; Path-gradient N (100.0%)	Equal-weight IG-N; IG-N; Path-gradient N (100.0%)
StackCube 170M	Q	Equal-weight IG-N; IG-N (70.0%)	Equal-weight IG-N; IG-N (85.0%)
StackCube 170M	N	Path-gradient N (98.3%)	Equal-weight IG-N; IG-N; Equal-weight IG-Q; IG-Q; Path-gradient N (100.0%)

Fixed-order raw means can be driven by unusual episodes. These three vision cells have a fixed-order raw mean above every attribution method. Rows retain that fixed winner, the raw and scaled attribution winners, and the additional task-trained center-out comparator. Counts in parentheses are complete episodes. All statistics first average calls within episodes; positive is the percentage of episode-mean advantages above zero. Scaling divides each call by its random-order standard deviation before averaging. These descriptive contrasts show sensitivity to aggregation, not significance or general method superiority.

Group, scoring, direction (episodes)	Method / order	Raw mean	Episode median	Positive (%)	Scaled mean
PickCube 170M, Q insertion (60)	Raster	7.06	−0.3126	10.0	−5.32
PickCube 170M, Q insertion (60)	Path-gradient N	−11.3	0.0857	81.7	0.993
PickCube 170M, N insertion (60)	Reverse raster	0.0354	0.0022	53.3	−0.335
PickCube 170M, N insertion (60)	Path-gradient N	−0.00388	0.1274	88.3	1.60
PickCube 1B, Q deletion (40)	Raster	1.31	0.0003	50.0	0.618
PickCube 1B, Q deletion (40)	Input difference	−0.572	0.0171	57.5	0.494
PickCube 1B, Q deletion (40)	Path-gradient N	−3.77	0.5819	87.5	8.55
PickCube 1B, Q deletion (40)	Center-out	0.813	0.2637	90.0	3.53

Separate numerical diagnostics

Table S5 distinguishes completeness, coordinate difference, ranking agreement and response-score change. It identifies each population and keeps the registered reference-qualification rule unchanged. Agreement between two numerical procedures is evidence of consistency, not proof that either equals the integral.

Table S5 | Reference qualification distinguishes agreement from accuracy. Eight contexts comprise four new PickCube contexts and four preserved contexts, one per task. Columns show median relative L1 differences (fractions). Qualification retains the registered three-way 0.001 tolerance and adaptive convergence condition: Q 15/24, N 5/24. Close adaptive/Gauss-Legendre agreement for N vision and language, with a larger trapezoid discrepancy, is consistent with unresolved endpoint behavior; it does not certify an exact integral. State retains larger independent-rule differences.

Target / input	Qualified	Adaptive vs Gauss	Adaptive vs trapezoid	Gauss vs trapezoid
Q vision	5/8	0.000000621	0.00000214	0.00000226
Q language	8/8	0.000000858	0.00000493	0.00000532
Q state	2/8	0.00635	0.0000179	0.00622
N vision	3/8	0.0000119	0.00171	0.00171
N language	1/8	0.00000230	0.00172	0.00172
N state	1/8	0.00426	0.000801	0.00499

Matched trapezoidal completeness residuals. These percentages use the same twelve pretrained-PickCube vision contexts in every row. Model precision changes; accumulation remains 32-bit. Figure 3 lower panels use eight of these twelve contexts because the 16-bit rule maps were available for only eight. Those eight are not the mixed-task qualification sample above. This comparison supports the precision statement without mixing the 120-, twelve- and eight-context subsets.

Target	Calculation	Contexts	64 intervals (%)	512 intervals (%)
Q	16-bit model	12	1.05	0.991
Q	32-bit model	12	0.0261	0.000321
N	16-bit model	12	4.76	4.46
N	32-bit model	12	1.54	0.192

Coordinate, ranking and perturbation diagnostics remain separate. Each comparison applies the old equal-weight 64-interval estimator to new pretrained-PickCube vision contexts and compares it with the corrected 16-bit, 512-interval trapezoid. Coordinate differences are relative L1 fractions. Ranking entries are minimum scored-order Spearman correlations; AUC changes concern deletion. Target and scoring response are separate in the first column.

Target / score	Contexts	Median L1	Min. rank correlation	Median / max AUC change	Changes above 0.01
Q/Q	120	0.0182	0.99931	0.00181 / 221	12/120
Q/N	120	0.0182	0.99931	0.00111 / 6.82	4/120
N/Q	120	0.0207	0.99957	0.00218 / 2.29	12/120
N/N	120	0.0207	0.99957	0.00151 / 0.527	7/120

The 21 preserved contexts also receive numerical checks. Entries are median completeness residuals in percent, with the same contexts across each row. These 21 contexts are all available members of the 24-context collection underlying the separate twelve-context completed diagnostic sample. They are not independent new robot runs. Each stratum has only two source episodes and supports no population inference. Old denotes the bf16 equal-weight 64-interval estimator. The two corrected columns use the 512-interval trapezoid and 32-bit accumulation, with 16-bit or 32-bit model calculations.

Preserved group	Input / target	Contexts	Old (%)	16-bit (%)	32-bit (%)
PickCube 170M	vision Q	4	1.25	2.89	0.0000524
PickCube 170M	vision N	4	9.07	7.41	0.0845
PickCube 170M	language Q	4	3.49	3.61	0.00103
PickCube 170M	language N	4	12.5	11.8	0.113
PickCube 170M	state Q	4	2.55	0.624	0.000422
PickCube 170M	state N	4	2.45	0.881	0.0441
PickCube 1B	vision Q	2	1.64	0.325	0.000217
PickCube 1B	vision N	2	4.25	2.62	0.0469
PickCube 1B	language Q	2	5.30	5.81	0.000831
PickCube 1B	language N	2	18.5	18.2	0.0712
PickCube 1B	state Q	2	53.1	8.55	2.86
PickCube 1B	state N	2	60.3	6.97	2.32
StackCube 170M	vision Q	4	2.92	1.67	0.000272
StackCube 170M	vision N	4	7.30	3.02	0.162
StackCube 170M	language Q	4	2.67	1.04	0.000527
StackCube 170M	language N	4	8.43	5.40	0.172
StackCube 170M	state Q	4	19.3	1.94	0.0157

Preserved group	Input / target	Contexts	Old (%)	16-bit (%)	32-bit (%)
StackCube 170M	state N	4	9.74	1.40	0.0420
StackCube 1B	vision Q	3	2.13	0.338	0.000206
StackCube 1B	vision N	3	2.75	0.985	0.0299
StackCube 1B	language Q	3	3.62	4.78	0.000383
StackCube 1B	language N	3	20.4	19.8	0.0989
StackCube 1B	state Q	3	0.466	0.296	0.000406
StackCube 1B	state N	3	2.01	1.24	0.109
Peg insertion 170M	vision Q	4	7.28	4.87	0.000710
Peg insertion 170M	vision N	4	9.11	6.49	0.250
Peg insertion 170M	language Q	4	3.15	0.639	0.000672
Peg insertion 170M	language N	4	7.71	5.89	0.137
Peg insertion 170M	state Q	4	3.34	2.93	0.00174
Peg insertion 170M	state N	4	8.04	6.70	0.0782
Object pickup 170M	vision Q	4	3.78	0.999	0.000131
Object pickup 170M	vision N	4	5.29	3.29	0.0361
Object pickup 170M	language Q	4	5.05	2.31	0.000702
Object pickup 170M	language N	4	9.57	8.00	0.248
Object pickup 170M	state Q	4	68.4	25.2	33.5
Object pickup 170M	state N	4	44.2	19.5	23.1

Separate original offline diagnostics: twelve contexts, six episodes, four pretrained tasks. This is the completed subset of the sealed collection, distinct from the later 21-context reruns. The model and summation use 32-bit arithmetic on stored 16-bit-valued weights and encoded image/language inputs; state is readapted and the self-reference recomputed in 32 bits. It is not a native 32-bit rollout. All 1,800 recorded equality checks passed, but repeatability did not establish quadrature accuracy. Entries are maxima for the two finest budgets: 256 versus 512 intervals for vision/language, 512 versus 1,024 for state. The finer estimate is a comparator, not ground truth. Any failed check counts contexts violating at least one recorded finer-pair criterion; these dependent checks do not estimate a failure frequency. IG and path-gradient columns are relative L1 differences. RMS is the largest pointwise response-curve change divided by baseline RMS, across both ranking types and directions. Extrema can come from different contexts. Two state contexts exceed 1% finest-budget completeness under each target. Figure S4 retains all adjacent-budget trajectories, including exact zeros.

Input / target	Any failed check	Max. IG L1 (%)	Max. path L1 (%)	Max. residual (%)	Max. RMS (%)
language N	0/12	0.402	0.425	0.315	0
language Q	0/12	0.00568	0.00537	0.00176	0

Input / target	Any failed check	Max. IG L1 (%)	Max. path L1 (%)	Max. residual (%)	RMS (%)
state N	2/12	20.8	22.0	6.65	18.6
state Q	2/12	32.1	34.4	11.4	1.64
vision N	4/12	1.23	1.19	0.288	7.36
vision Q	1/12	2.05	2.02	0.124	0

Success splits and additional tasks

Table S6 reports medians, episode-resampling intervals, overshoots and extreme-tail rates. Different sample sizes affect observed minima. Success and failure are observed outcomes, not randomized treatments, and differences between the saved and matched groups cannot be attributed to success alone.

Table S6 | Vision response changes by success and task. Counts are episodes / calls. All intervals resample whole episodes; the new 1B intervals are retrospective. Overshoot and extreme-tail counts are call counts. Unequal sample sizes affect extrema, and episode dependence remains. The saved 1B rows use the 64-interval equal-weight rule, the new 1B rows the 512-interval trapezoid, and the additional tasks the 128-interval equal-weight rule. All use target Q. All additional-task episodes failed.

Group / direction	Episodes / calls	Median [95% interval]	Overshoot	Below -1	Min.
Saved 1B success insertion	15 / 74	0.0653 [0.0522, 0.0755]	18/74	1/74	−2.35
Saved 1B failure insertion	45 / 1125	0.0868 [0.0775, 0.0939]	326/1125	8/1125	−57.7
New 1B success insertion	31 / 62	0.0616 [0.0486, 0.0744]	13/62	2/62	−89.9
New 1B failure insertion	9 / 18	0.0913 [0.0661, 0.104]	7/18	0/18	0.0465
Peg insertion insertion	16 / 400	0.0990 [0.0705, 0.125]	276/400	34/400	−212
Object pickup insertion	16 / 400	0.0954 [0.0815, 0.110]	118/400	6/400	−35.4
Saved 1B success deletion	15 / 74	0.0747 [0.0422, 0.0995]	31/74	0/74	−0.347
Saved 1B failure deletion	45 / 1125	0.1035 [0.0908, 0.115]	355/1125	4/1125	−74.5
New 1B success deletion	31 / 62	0.0212 [0.00230, 0.0500]	41/62	4/62	−57.7
New 1B failure deletion	9 / 18	−0.0086 [−0.177, 0.0608]	13/18	0/18	−0.639
Peg insertion deletion	16 / 400	0.1152 [0.0745, 0.145]	166/400	3/400	−24.7
Object pickup deletion	16 / 400	0.1405 [0.123, 0.155]	98/400	5/400	−4.37

Saved task-trained language results, separate by seed and baseline. Seed 42 uses the RDT developers' precomputed instruction embedding and a zeroed baseline; seeds 142 and 242 use generated T5-XXL embeddings and encoded PAD/EOS baselines. The separate seed-42 rerun uses generated embeddings and an encoded baseline and is a sensitivity sample, excluded from the 60 primary episodes. All use target Q. Intervals resample whole episodes. The single successful seed-142 episode cannot support an episode-bootstrap interval. Overshoots are percentages of calls.

Seed, baseline	Outcome / direction	Episodes / calls	Median [95% interval]		Overshoot (%)	Min.
42, zeroed	success insertion	9 / 45	−0.0973	[−0.134, 0.0104]	93.3	−10.8
42, zeroed	success deletion	9 / 45	0.0384	[−0.0508, 0.0932]	80.0	−1.29
42, zeroed	failure insertion	11 / 275	0.0186	[−0.0521, 0.0772]	80.0	−145
42, zeroed	failure deletion	11 / 275	0.1000	[0.0154, 0.147]	56.7	−1.98
142, encoded	success insertion	1 / 5	−0.0764	Not estimable	100	−0.869
142, encoded	success deletion	1 / 5	−0.3160	Not estimable	100	−2.52
142, encoded	failure insertion	19 / 475	0.0533	[−0.00712, 0.0840]	85.3	−95.9
142, encoded	failure deletion	19 / 475	0.1095	[0.0854, 0.131]	59.8	−2.83
242, encoded	success insertion	5 / 24	−0.0460	[−0.171, 0.103]	91.7	−5.32
242, encoded	success deletion	5 / 24	0.0071	[−0.0694, 0.110]	83.3	−1.01
242, encoded	failure insertion	15 / 375	0.0640	[0.00874, 0.0863]	88.5	−109
242, encoded	failure deletion	15 / 375	0.0492	[−0.0132, 0.0915]	72.8	−549
42 rerun, encoded	success insertion	8 / 34	−0.0431	[−0.253, 0.0728]	97.1	−7.10
42 rerun, encoded	success deletion	8 / 34	−0.0732	[−0.197, 0.0628]	85.3	−2.00
42 rerun, encoded	failure insertion	12 / 300	0.0578	[−0.240, 0.0965]	82.7	−158
42 rerun, encoded	failure deletion	12 / 300	0.0825	[−0.0696, 0.131]	69.0	−281

Retrospective winner stability

Table S7 uses the seven attribution methods on common complete episodes, with the same resampled episodes for Q and N. Each of 10,000 draws, seed 930011, produces the same unique winner, different unique winners or an undetermined tie. Raw means average per-episode advantages, scaled means first divide each context by the standard deviation of its random-order scores, and episode medians take the median of per-episode means. These frequencies describe sensitivity to resampling the recorded episodes. They are not probabilities about a population's true preferred method and were not part of the registered primary analysis.

Table S7 | Retrospective stability of vision-deletion choices. Entries are percentages of 10,000 paired whole-episode bootstrap draws, seed 930011. Every draw uses the same episodes for both responses and all seven methods. Exact ties are retained separately. These are resampling frequencies for these episodes, not probabilities of a true population winner. The median criterion uses per-episode mean raw advantages.

Group	Criterion	Same winner (%)	Different winners (%)	Any tie (%)
PickCube 170M	Raw mean	20.37	79.63	0.00
PickCube 170M	Scaled mean	90.02	9.98	0.00
PickCube 170M	Episode median	36.71	63.29	0.00
PickCube 1B	Raw mean	35.87	64.13	0.00
PickCube 1B	Scaled mean	98.48	1.52	0.00
PickCube 1B	Episode median	54.92	45.08	0.00
StackCube 170M	Raw mean	69.95	30.05	0.00
StackCube 170M	Scaled mean	60.47	39.53	0.00
StackCube 170M	Episode median	37.13	62.87	0.00

Numerical estimator definitions

Table S8 specifies model precision, accumulation, integration rule and node budgets for each numerical variant.

Table S8 | Numerical variants and their purpose. The old rule generated saved attributions and is reapplied to new contexts; these are distinct data sources. Numerical agreement is a diagnostic, not an exactness guarantee.

Variant	Calculation / sum	Rule and evaluations	Purpose
Old equal-weight rule	bf16 / bf16	65 uniformly spaced nodes, equal weights	Recover the old implementation

Variant	Calculation / sum	Rule and evaluations	Purpose
Reduced-precision check	bf16 / fp32	513 shared nodes: equal-weight/trapezoid through 512 intervals; midpoint through 256	Separate summation from model precision
32-bit check	fp32 / fp32	The same 513 nodes and nested rules	Matched precision comparison
Gauss-Legendre	fp32 / fp32	32, 64 or 128 interior nodes	Independent rule comparison
Adaptive Gauss-Kronrod	fp32 / fp32	Adaptive interior nodes	Independent convergence check

Additional visual evidence

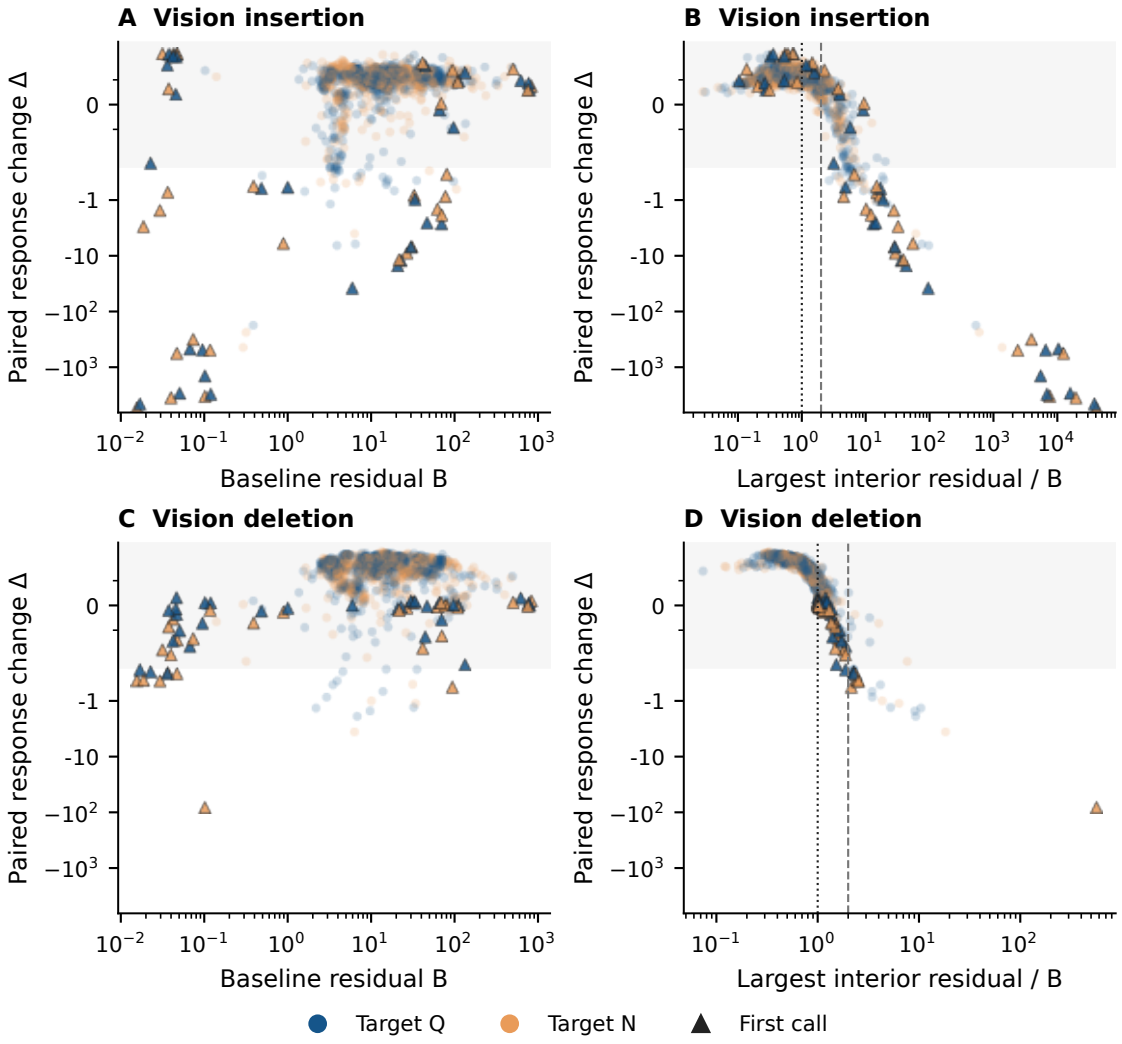


Figure S1 | Extreme response differences occur with small baseline distances and large residual ratios. The 3,000 plotted vision calls retain their recorded case identity: insertion in A/B, deletion in C/D, with first calls distinguished from later calls. Baseline squared distance is B; the interior residual ratio is the largest sampled r/B. Lines at ratios one and two mark overshoot and a conservative threshold. Symmetric-log response axes are linear for $|\Delta| \leq 0.25$ and compress larger magnitudes while preserving sign. Mathematical threshold constraints are described above; the display supports a descriptive, unproven mechanism.

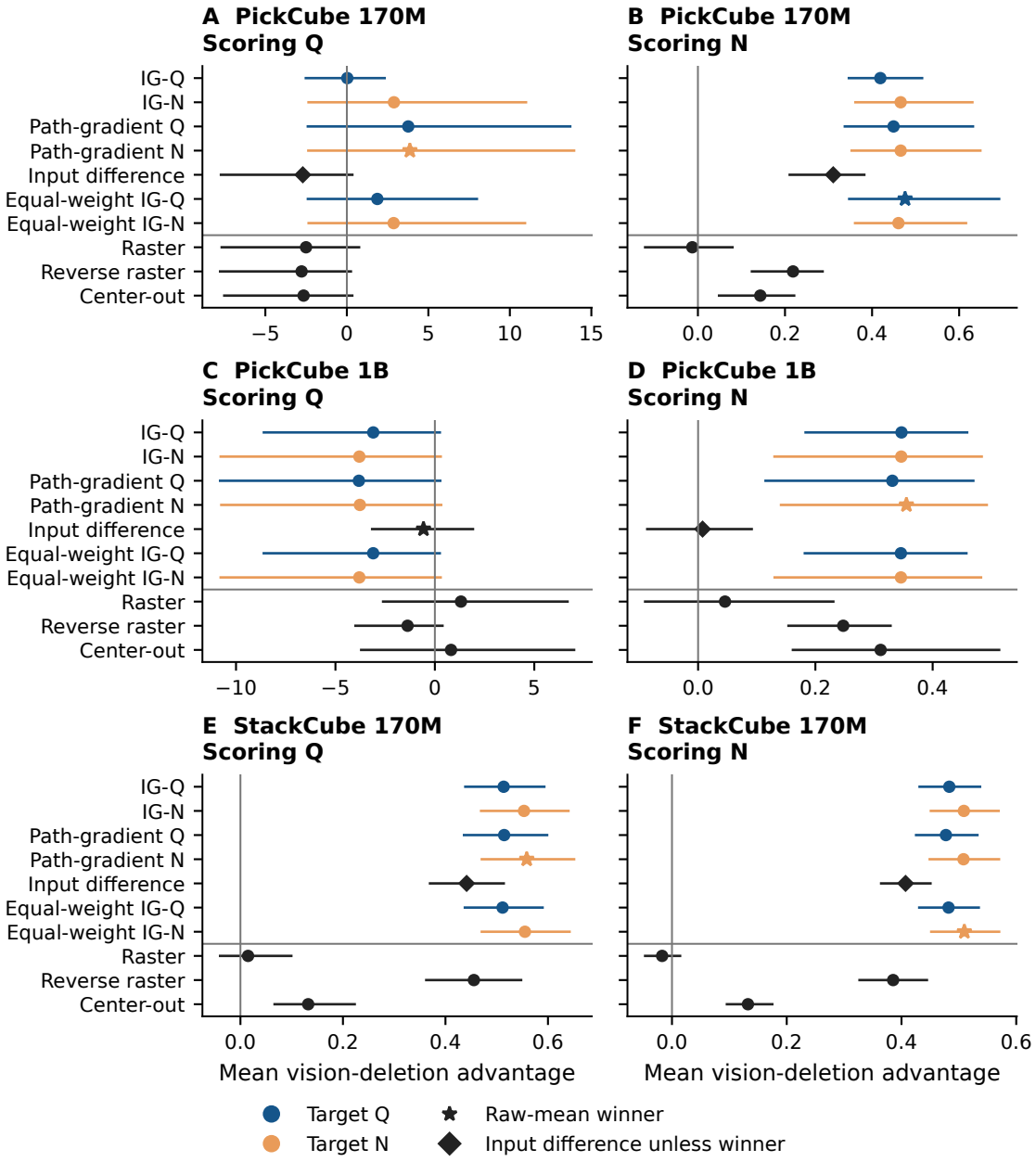


Figure S2 | Matched vision-deletion scores depend on the response and the recorded group. The three groups are pretrained PickCube, task-trained PickCube and pretrained StackCube, each evaluated under Q and N. Zero is the response-matched random mean; positive advantage favors the ranking. Intervals resample whole episodes. Separate intervals do not establish paired method differences; Table S3 gives those adjusted comparisons. Stars mark raw-mean winners, diamonds input difference when it is not the winner. Blue/orange distinguish Q/N targets; black marks input difference and fixed controls. IG and path gradients use the 512-interval trapezoid; equal-weight IG uses the original 64-interval estimator. A divider separates fixed controls. Wider Q panels reflect extreme tails.

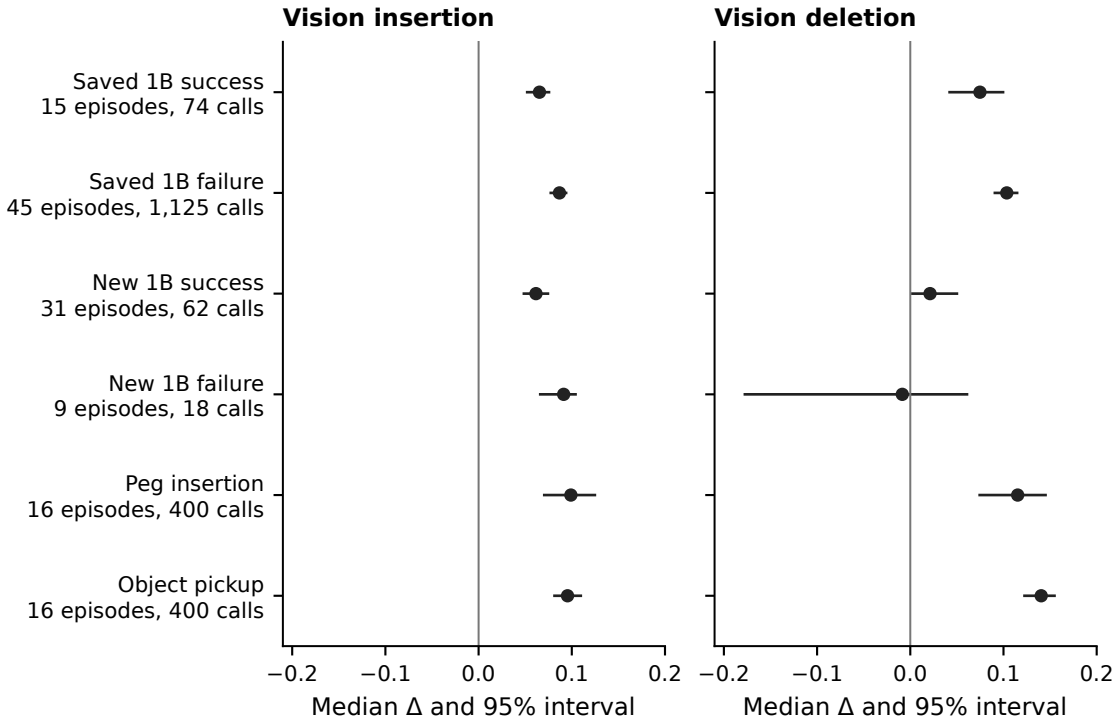


Figure S3 | Positive median response differences do not remove overshoots. Vision insertion and deletion summaries retain saved task-trained PickCube, the two saved additional tasks, and the new task-trained PickCube group separately. Counts and 95% whole-episode bootstrap intervals identify each population. The failed new group's deletion median is uncertain rather than evidence for a negative population effect. Table S6 supplies the full intervals and tail shares.

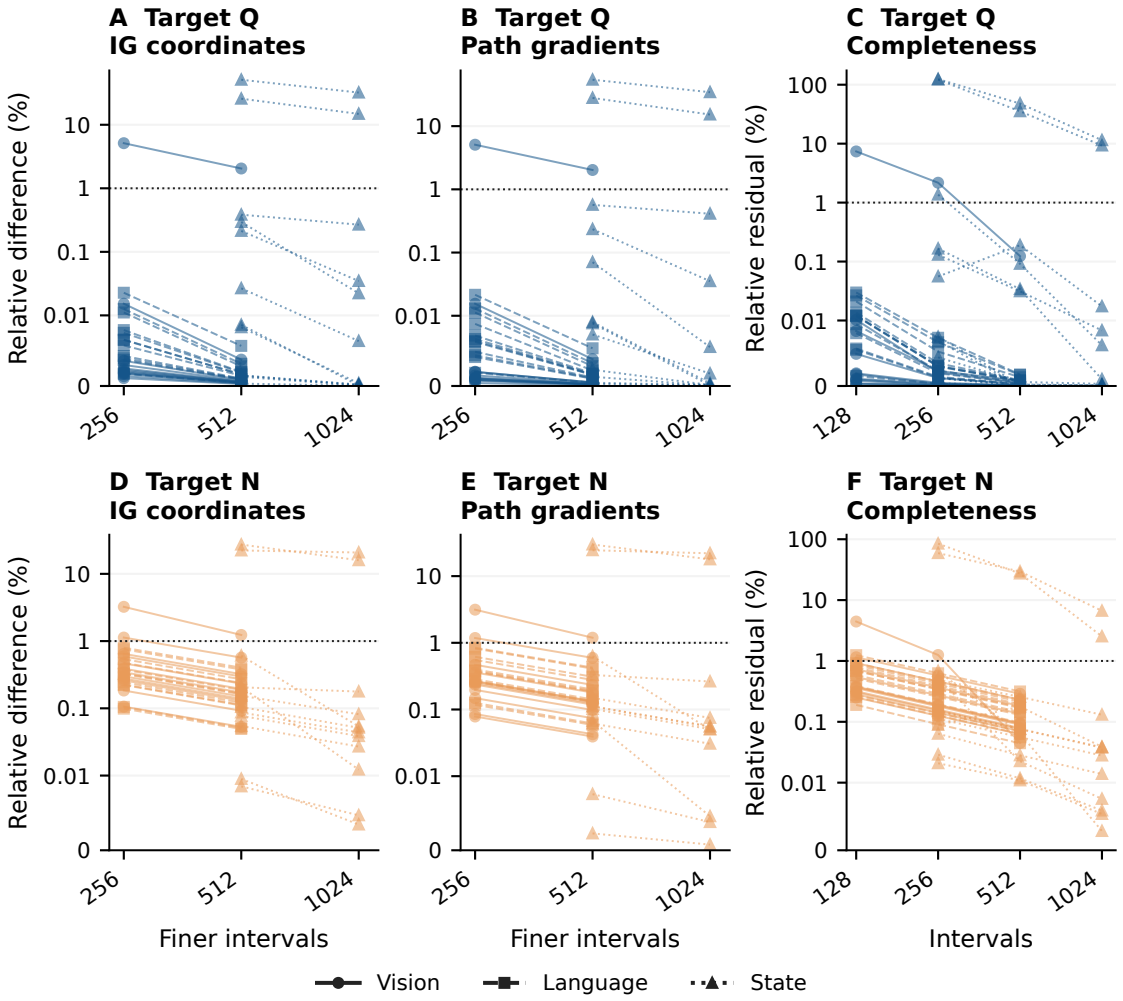


Figure S4 | Repeated function agreement does not establish attribution convergence. The original 32-bit offline diagnostic sample contains twelve contexts from six episodes across four pretrained-model tasks. **A), D)** Relative L1 changes in IG coordinates between adjacent integration budgets; **B), E)** corresponding path-gradient changes. The horizontal coordinate is the finer budget in each m versus $2m$ comparison. **C), F)** Completeness residuals at each budget. All vertical quantities are percentages; symmetric-log axes retain exact zeros and are linear through 0.01%. Dotted horizontal lines mark the study-specific 1% criterion. Blue/orange denote Q/N targets; circles, squares and triangles denote vision, language and state. The finest pairs use 256/512 intervals for vision and language, 512/1,024 for state. The finer estimate is a comparator, not truth. Comparisons are dependent and do not estimate numerical-failure frequency. Supplementary Table S5 gives maxima and the 1,800 recorded repeat checks.